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Investigation report

Why MAU grew 20% but revenue stayed flat

Monthly active users increased about 20% quarter over quarter, but revenue stayed nearly flat. What explains the difference?

Data

PocketToons synthetic data
Q2 2026 (Apr–Jun) vs Q3 2026 (Jul–Sep), UTC

Snapshot

sha256 8d39e78b…790b
Build 3a0dbed2

Run

Independent run · v2, corrected after peer review
2026-11-09
21 logged checks

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MAU vs revenue, Q2–Q3 2026

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How to read this report

- **Check N** = one logged database query. Every number points to the check that produced it. The appendix lists all 21 checks in plain words.
- **Data layers:** Gold = pre-built monthly tables, Silver = cleaned user-level tables, Bronze = raw records.
- **Signup groups:** Q2 signups joined in April–June; Q3 signups joined in July–September. 30-day results leave out September signups, because their 30 days run past the end of the data (30 September).
- **Patterns, not proof:** where the report says a pattern “suggests” something, the cause has not been proven.

Terms used in this report

Term	Meaning
MAU	Monthly active users: distinct signed-in people with at least one session in the month. The quarter figure is the average of its three months.
Net revenue	Successful charges minus refunds, counted on the day the payment succeeds. App-store fees and tax are not deducted.
Signup cohort	Everyone who created an account in the same calendar month. Example: the July cohort = all accounts created 1–31 July. We follow each cohort for its first 30 days.
Paywall	The screen that asks a user to subscribe or buy before watching premium content.
Paywall reach	Share of a signup cohort who saw at least one paywall within 30 days of their own signup time.
Paid within 30 days	Share of a signup cohort with a successful payment within 30 days of signing up (30-day paid conversion).
Unidentified session	A session we can't link to a signed-in user, so it isn't counted in MAU.
ARPU / ARPPU	Average revenue per active user / per paying user.

01. Chapter

The answer

01

Recent signup cohorts show much lower recorded paywall reach and paid conversion. This suggests a problem early in the path to paying, but the data doesn't show whether the cause is user behaviour, product design, tracking or something else.

+20.0%
Average MAU, Q3 vs Q2
9,016 → 10,819

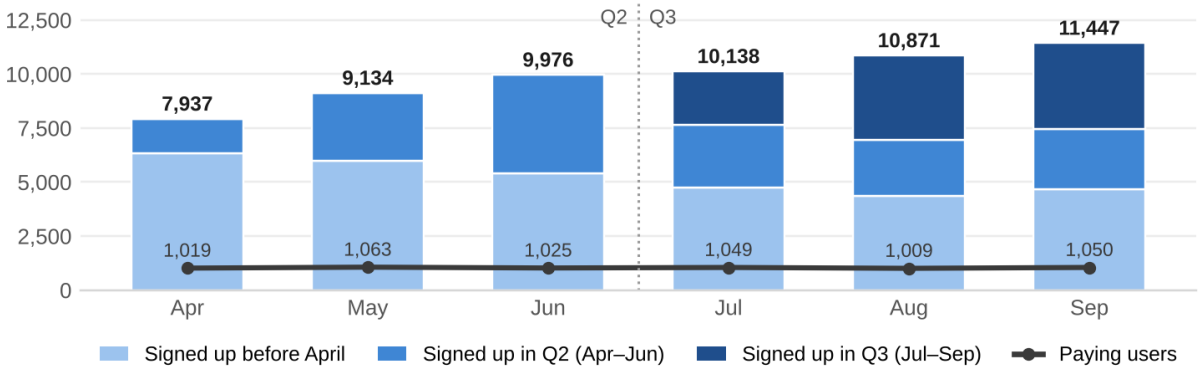
+0.07%
Net revenue, Q3 vs Q2
$30,145 → $30,167

~1,036
Paying users per month in both quarters

0.5%
Paid within 30 days, Jul–Aug signups (May–Jun: 1.9%)

Active users grew every month; paying users didn't

Monthly active users split by when they signed up, and paying users per month



Source: Silver daily activity and user profiles (check 12); paying users from the Gold monthly KPI table (check 1). The three groups don't overlap and add up to MAU.

- **Who the new users are:** all of the net MAU growth came from Q3 signups (+3,457 a month on average), mostly from paid social ads. Users who joined earlier fell by 1,654 (checks 6, 12).
- **What is different for them:** they watch about as much as earlier signups (about 3.6 views in the first 30 days), but fewer saw a paywall (1.0% in July, 0.5% in August, versus 4.4% in May and 2.7% in June). Fewer paid within 30 days: 0.5% of July–August signups versus 1.9% of May–June signups (checks 7, 17).
- **Why revenue is flat:** revenue = paying users × revenue per paying user, and both stayed flat (about 1,036 payers paying about $9.70 each) (checks 1, 5).

Not yet known: why paywall reach and conversion fell, and whether showing more paywalls would raise conversion. Two data questions are also open: how many people sit behind unidentified sessions, and why checkouts from Q2 signups stopped reaching payment in August–September (chapter 05).

02. Chapter

How we investigated

02

What data we used

Item	Detail
Database	pockettoons.sqlite, opened read-only and not changed (SHA-256 8d39e78b…790b)
Build	Built 2026-11-09 11:16 UTC; all 122 automatic consistency checks passed
Coverage	1 April – 30 September 2026 (UTC)
Run	Independent: no answer keys, guides or earlier reports were read. 16 checks in the original run, plus 5 follow-up checks after peer review.
Version	v2, corrected after peer review. The original agent output (v1) is kept unchanged for comparison.

Investigation path

How to read: Gold pre-built monthly tables Silver cleaned user-level tables Bronze raw records
Supported evidence agrees **Rejected** evidence disagrees **Inconclusive** evidence can't decide

Start

Can we trust this database, and is the question's premise true?

Passed Build log **Database health**
One successful build; 122 of 122 consistency checks passed; 20 Bronze views and 18 Silver/Gold tables present.

Passed Gold **MAU up 20%, revenue flat?** (check 1–2)
Yes: average MAU +20.0%, net revenue +0.07%.

→ **MAU and revenue disagree, so look at who the active users are and who pays.**

Round 1

Where did the extra users come from, and did anything change on the money side? Gold tables answer this directly.

Supported Gold **H1 Growth comes from new users who don't pay** (check 3, 4, 6)
Growth is in new users, mostly paid social; subscribers flat near 1,000.

Rejected Gold **H2 Price, mix, refunds or failed payments offset growth** (check 5)
Price $9.99 throughout; refunds under $40 a quarter; failure rate 5–6% in both quarters.

Rejected Gold **H3 Growth limited to one platform or country** (check 6)
Growth on every platform and in every country.

→ **Growth is new users who don't pay, so test why new users don't convert.**

Round 2

Why don't new users convert? Gold has cohort totals; splitting by source needs Silver.

Rejected Silver **H4 Only paid social users convert worse** (check 7, 8)
Conversion fell to about 0.5% in every source, organic included.

Supported Gold **H5 Paywall views didn't grow with MAU** (check 9)
About 1,000–1,050 paywall viewers a month from May on, while MAU grew.

Pattern supported Silver **H6 New users rarely reach the paywall** (check 10, 12, 14)
Pattern confirmed. Whether it causes lower conversion is not tested.

→ **Rule out content changes and measurement problems before trusting the pattern.**

Round 3

Could content or measurement explain the pattern? History and raw events need Bronze.

Rejected Bronze **H7 Premium episodes became free** (check 13)
Only 1 episode changed tier.

Inconclusive Silver **H8 Unidentified sessions distort MAU** (check 15)
Share is 6.5–7.6% and varies; how many people they represent is unknown.

Inconclusive Bronze **H9 Paywall events stopped being tracked** (check 16)
Payments fall with paywall views, but that can't prove paywall events are complete.

→ Budget of 3 rounds used. Pattern measured, cause not proven: report it and recommend a test and a tracking audit.

Review follow-up

Checks added after peer review of v1.

Passed Silver **First-30-day window with exact time limits** (check 17)
Same results as check 14; no paywall views before signup. Viewing uses whole UTC days.

Passed Silver **Size of the 13 corrected payments** (check 18)
$71.92 in Q2 and $49.95 in Q3; with all of them removed, the QoQ change is +0.15%.

Inconclusive Silver **Q2 signups' checkouts in Aug–Sep** (check 19–21)
183 checkouts, none reached a payment attempt. Cause unknown.

03. Chapter

What happened

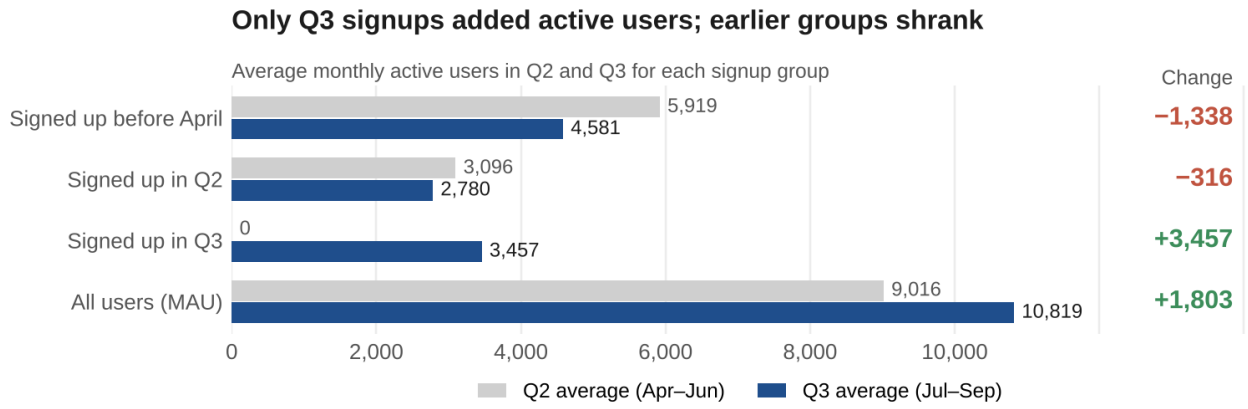
03

The movement

Measure	Q2 (Apr–Jun)	Q3 (Jul–Sep)	Change
Average monthly active users (MAU)	9,016	10,819	+20.0%
Net revenue (sum of 3 months)	$30,145.00	$30,166.97	+0.07% (+$21.97)
Average paying users per month	1,035.7	1,036.0	about 0%

Source: Gold monthly KPI table (checks 1–2). Revenue per active user (ARPU) fell from $1.24 in April to $0.89 in September: more people use the product, but the number of payers hasn't changed. The month-by-month view is the chart in chapter 01.

Who drove it



Source: Silver daily activity and user profiles (check 12); totals match the Gold monthly KPI table.

How the total change adds up

The three groups don't overlap, so their changes add up to the total change in MAU:

9,016 (Q2 MAU) − 1,338 (signed up before April) − 316 (signed up in Q2) + 3,457 (signed up in Q3) = 10,819 (Q3 MAU)

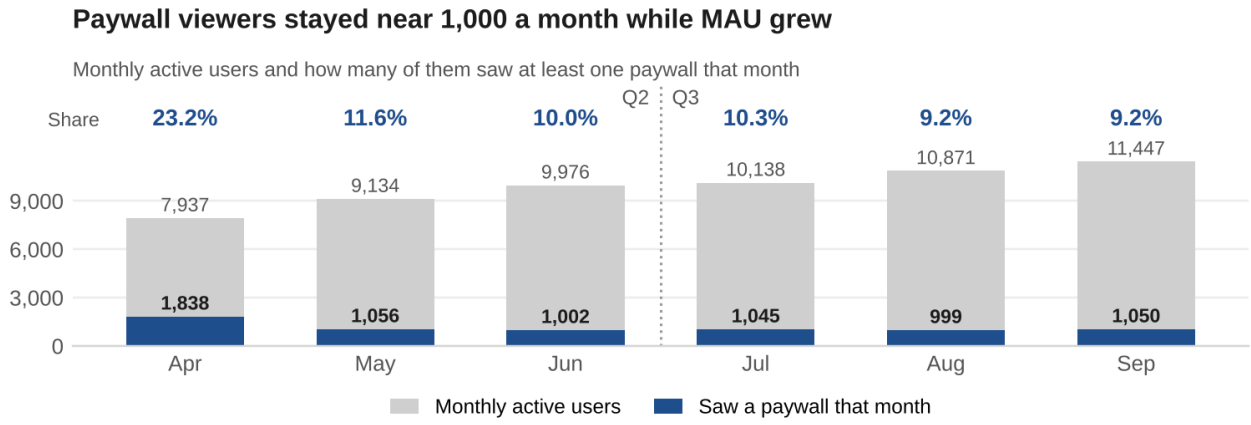
Q3 signups had no activity in Q2 because they didn't exist yet, so their whole Q3 average counts as growth.

- **By acquisition source** (check 6): paid social +1,841, paid search +92, unknown +25, organic −156. Paid social was 24% of May signups and 64% of July–August signups.
- **Revenue side** (check 5): 99% of revenue is the $9.99 monthly subscription. About 1,000 subscribers pay each month; roughly 10% cancel each month and about as many join, so the base stays flat.

04. Chapter

Why it happened

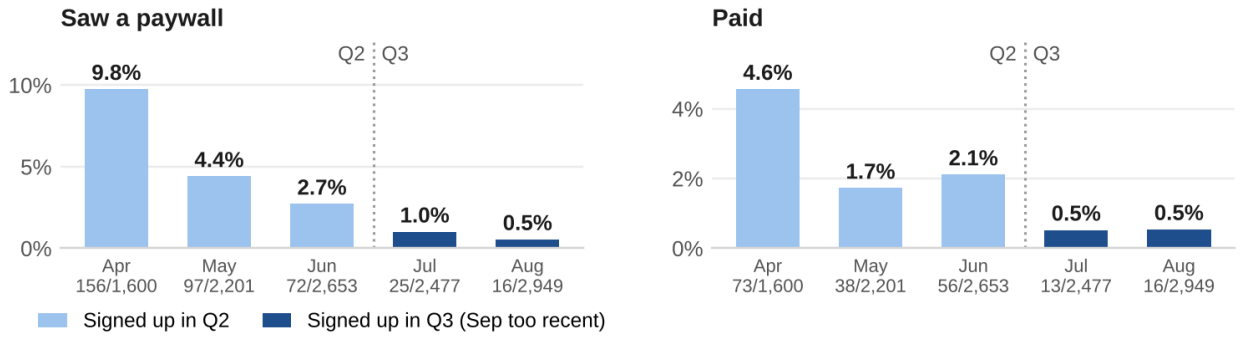
04



Source: Gold paywall funnel table (check 9). April is high because the data starts with no subscribers, so many users subscribe that month.

New users' first 30 days: fewer saw a paywall, fewer paid

Each bar = users in that signup month who did it within 30 days ÷ all users who signed up that month (counts under each month)



Source: Silver user profiles, paywall funnel and payments (checks 7, 17). Exact 30-day windows from each user's signup time.

The maths behind the bars

Saw a paywall % = signups who saw at least one paywall within 30 days ÷ all signups that month. Example: April 156 ÷ 1,600 = 9.8%.

Paid % = signups with a successful payment within 30 days ÷ all signups that month. Example: July 13 ÷ 2,477 = 0.5%.

Comparison used: May–June 94 ÷ 4,854 = 1.9% versus July–August 29 ÷ 5,426 = 0.5%. April is left out because the data starts with no subscribers; September is left out because its 30 days end after the data does.

Every 30-day payer also saw a paywall (check 17). That shows the two go together, not that the paywall causes payment.

All 9 explanations tested (the investigation path is in chapter 02):

#	Possible explanation	Status	What the data showed
H1	Growth comes from new users who don't pay	**Supported**	Paid social users +1,841 of the +1,803 total; subscribers flat near 1,000 (checks 4, 6, 12)
H2	Price, product mix, refunds or failed payments offset growth	**Rejected**	$9.99 price; refunds under $40 a quarter; failure rate 5–6% in both quarters (check 5)
H3	Growth is limited to one platform or country	**Rejected**	Growth on every platform and in every country (check 6)
H4	Only paid social users convert worse	**Rejected**	About 0.5% in every source for July–August signups (check 8)
H5	Paywall views didn't grow with MAU	**Supported**	About 1,000–1,050 viewers a month from May; share of MAU 11.6% → 9.2% (check 9)
H6	New users rarely reach the paywall	**Pattern supported**	Reach 4.4% (May) → 0.5% (Aug) with similar viewing. Causal effect on conversion not tested (checks 14, 17)
H7	Premium episodes became free	**Rejected**	Only 1 episode changed tier (check 13)
H8	Unidentified sessions distort MAU	**Inconclusive**	6.5–7.6% of sessions, varying by month; the number of people they represent is unknown (check 15)
H9	Paywall events stopped being tracked	**Inconclusive**	Doesn't rule out missing paywall events; needs an instrumentation audit (checks 16, 21)

Scenario, not a forecast: if July–August signups (5,426) had paid at the May–June rate (1.9%), about 105 would have paid instead of 29. That is about 76 more subscribers × $9.99 ≈ $760 a month, around 7.6% of Q3's average monthly net revenue ($10,056). It assumes the rate is reachable, the price stays at $9.99 and these subscribers renew.

05. Chapter

Data quality and limits

05

Data quality

These issues don't invalidate the calculations on the current data snapshot. But their effect on how the history is read, and on the size of the movement, is not fully measured.

Issue	Effect on the answer	Status
Data starts with 0 subscribers on 1 April	April paywall and conversion figures are inflated while the base forms	Handled: May–June compared with July–August
6.5–7.6% of sessions can't be linked to a user (share varies by month)	MAU leaves these out. How many people they represent, and whether that changes the Q2 vs Q3 comparison, is unknown	Open (H8, check 15)
13 raw payment amounts corrected after review; review dates unknown	Current corrected amounts are used. If all 13 were removed: Q2 −$71.92, Q3 −$49.95, QoQ change +0.15%. When the corrections became available is unknown	Measured (check 18); history unknown
Q2 signups' checkouts in Aug–Sep never reached payment	183 checkouts, no payment attempt and no purchase. Other groups also abandon checkouts often (61–85% for pre-April users), but 100% is unusual. Could be behaviour, a product issue or broken event linking, and it bears on the paywall explanation	Unresolved (checks 19–21)
16–27 checkouts a month without a captured paywall link	Paywall-to-purchase rates may be slightly understated	Counted, not guessed

Limitations

- The data shows where new users stop, not why. Product release and campaign data aren't available.
- PocketToons is synthetic data. Patterns here are associations, not proven causes.
- First-30-day viewing figures use whole UTC days, so the first and last day are approximate. Paywall and payment figures use exact times.
- Not checked: marketing spend vs new payers, which titles lead to paywalls, and subscriber churn by signup month.

06. Chapter

What to do next

06

01 Find what changed for new users around 1 July. Check onboarding, paywall placement and the paid social landing experience. *Success check:* 30-day paywall reach returns towards the May–June level (2.7–4.4%).

02 Audit paywall and checkout tracking. Confirm paywall events are complete, and explain why Q2 signups' checkouts stopped reaching payment in August–September. This decides whether H9 can be closed.

03 Track two early-warning metrics next to MAU: new-user paywall reach within 30 days, and 30-day paid conversion by source.

04 Judge paid social on payers, not active users. July–August paid social signups produced 18 payers. Compare spend with new payers before increasing the budget.

05 Test the cause with an A/B experiment: show a paywall prompt to new free users (for example, at the free-episode limit) to learn whether more exposure actually raises conversion.

Appendix

Evidence appendix

A

Each check ran through the repo's read-only evidence tool and has its own log: the SQL, data layer, metric definition version, row count and a truncation flag. No result was truncated. Original run: 11:48–11:51 UTC on 2026-11-09. Review checks 17–21 ran later that day. All logs are in the evidence bundle.

Checks	What they asked	Layer	Log files
1–2	Monthly MAU, revenue, payers; quarter comparison	Gold	q01–q02
3–7	New/returning users, subscriber status, revenue by type, segments, cohorts	Gold	q03–q07
8	30-day conversion by source and cohort	Silver	q08
9, 11	Paywall funnel by month; viewing and free-episode limit	Gold	q09, q11
10, 12, 14	Paywall viewers and first-30-day behaviour by signup group	Silver	q10, q12, q14
13, 16	Episode tier changes; blocked playback vs paywall events	Bronze	q13, q16
15	Share of sessions not linked to a user	Silver	q15
17–18 (review)	Check 14 with exact time limits; size of the 13 corrected payments	Silver	q17–q18
19–21 (review)	Q2 signups' checkouts and payment attempts; checkouts without payment	Silver	q19–q21

Metric definitions used (versions from the project's metric registry, for audit): MAU 1.1, revenue group 1.1, funnel and lifecycle 1.0, engagement 1.0, growth accounting 1.0. Database build 3a0dbed21d2f45958aefa2887a0cadd8.